

Experiment 1: Probability Theory

Aim

To calculate the probability of each class (Malignant and Benign) using the Breast Cancer Wisconsin Diagnostic Dataset and predict the class of a new patient.

Theory

Probability theory is used to determine the likelihood of an event. In machine learning, class probability is computed as

$$P(\text{Class}) = \text{Number of instances of the class} / \text{Total number of instances}$$

The class with the highest probability is considered the predicted class.

Algorithm

1. Import dataset.
2. Read dataset using pandas.
3. Count Malignant and Benign samples.
4. Calculate class probabilities.
5. Accept a new sample.
6. Predict class based on maximum probability.
7. Display results.

Sample Dataset

ID Radius Texture Diagnosis

1	17.99	10.38	M
2	20.57	17.77	M
3	19.69	21.25	M
4	11.42	20.38	B
5	20.29	14.34	M
6	12.45	15.70	B

ID Radius Texture Diagnosis

7	18.25	19.98	M
8	13.71	20.83	B
9	13.00	21.82	B
10	12.46	24.04	B

Python Code

```
import pandas as pd
```

```
data = {  
    'Diagnosis':['M','M','M','B','M','B','M','B','B','B']  
}
```

```
df = pd.DataFrame(data)
```

```
malignant = (df['Diagnosis']=='M').sum()
```

```
benign = (df['Diagnosis']=='B').sum()
```

```
total = len(df)
```

```
p_m = malignant/total
```

```
p_b = benign/total
```

```
print("Total Samples:",total)
```

```
print("Malignant:",malignant)
```

```
print("Benign:",benign)
```

```
print("Probability of Malignant =",round(p_m,2))
```

```
print("Probability of Benign =",round(p_b,2))
```

```
if p_m>p_b:
```

```
    print("Prediction : Malignant")
```

```
else:
```

```
    print("Prediction : Benign")
```

Sample Output

Total Samples: 10

Malignant: 5

Benign: 5

Probability of Malignant = 0.50

Probability of Benign = 0.50

Prediction : Benign

Result

The probability of each class was calculated successfully and the new instance was classified using the highest probability.